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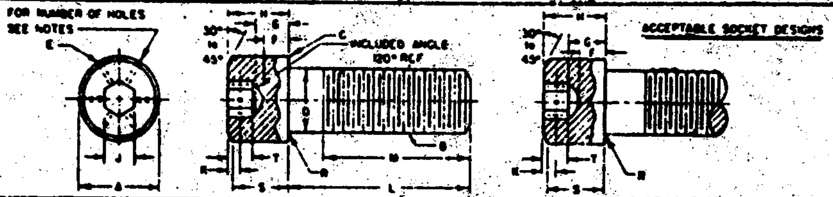

NOMINAL SIZE		#4 (.112)	#6 (.138)	#8 (.164)	#10 (.190)	1/4
– THREADS PER INCH		40UNC-3A	32UNC-3A	32UNC-3A	24UNC-3A	20UNC-3A
D- BODY DIAMETER	MAX	.1120	.1380	.1640	.1900	.2500
	MIN	.1075	.1329	.1585	.1840	[illegible]
A- HEAD DIAMETER	MAX	.183	.226	.270	.312	.375
	MIN	.176	.218	.262	.303	[illegible]
H- HEAD HEIGHT	MAX	.112	.138	.164	.190	.250
	MIN	.108	.134	.159	.185	[illegible]
F- HOLE DEPTH	MIN	.009	.013	.011	.016	[illegible]
S- HEAD SIDE HEIGHT	MIN	.101	.126	.148	.171	[illegible]
J- SOCKET WIDTH ACROSS FLATS	MAX	.0952	.1111	.1426	.1587	[illegible]
	MIN	.0937	.1094	.1406	.1562	[illegible]
T- KEY ENGAGEMENT	MIN	.051	[illegible]	.077	[illegible]	[illegible]
G- WALL THICKNESS	MIN	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
– RADIUS OR CHAMFER	MAX	.005	.005	.005	.005	[illegible]
R- RADIUS	MAX	.009	.010	.012	.014	[illegible]
	MIN	.005	.006	.007	.009	[illegible]
K- HOLE LOCATION	MAX	.040	.050	.060	[illegible]	[illegible]
	MIN	.026	.035	.040	.045	[illegible]
E- HOLE DIAMETER	MAX	.039	.039	.050	.050	[illegible]
	MIN	.033	.033	[illegible]	[illegible]	[illegible]
B- MINIMUM THREAD LENGTH		.750	.750	.875	.875	1.000
RECOMMENDED WIRE DIAMETER	MAX	.028	.028	.040	.040	.040
	MIN	.020	.020	.032	.032	.032
*DRILL HOLE ALIGNMENT PLUG		.025	.025	.032	.032	.036
**MINIMUM BREAKING STRENGTH		990 LB	1,400 LB	2,170 LB	2,710 LB	[illegible] LB

L- LENGTH	PART NO.	PART NO.	PART NO.	PART NO.	PART NO.
1/4 [illegible] 7/8 1 [illegible] 2 [illegible]	MS 24676 -1 -2 -3 -4 MS 24676 -5	MS 24676 -6 -7 -8 -9 -10 -11 MS 24676 -12	MS 24676 -13 -14 -15 -16 -17 -18 -19 -20 MS 24676 -21	MS 24676 -22 [illegible] MS 24676 -31	MS 24676 -32 -33 [illegible] MS 24676 [illegible]

Ⓑ CANCELLED
AFTER 7 JULY 1975 USE MS 24677
SEE SHEET 4

Ⓑ FOR CHANGES SEE SHEETS 1, 2, 3, 4

APPROVED 7 APRIL 1960 REVISED Ⓐ 27 JAN 1961 Ⓑ 7 JULY 1975

P.A. [illegible]
Other Cust
C/D
ARS

PROCUREMENT SPECIFICATION
[illegible]

TITLE
SCREW, CAP - SOCKET HEAD, HEXAGON, DRILLED
ALLOY STEEL, [illegible] PLATED
[illegible]

SUPERSEDES

MILITARY STANDARD
MS24676
SHEET 1 OF 4

FED SUP CLASS
[illegible]

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NOMINAL SIZE		5/16	3/8	7/16	1/2	5/8
T- THREADS PER INCH		18UNC-3A	16UNC-3A	14UNC-3A	13UNC-3A	11UNC-3A
D- BODY DIAMETER*	MAX MIN	.3125 .3053	.3750 .3678	.4375 .4294	.5000 .4919	.6250 .6163
A-HEAD DIAMETER	MAX MIN	.468 .457	.562 .550	.656 .642	.750 .735	.937 .921
H- HEAD HEIGHT	MAX MIN	.312 .306	.375 .368	.437 .430	.500 .492	.625 .615
S- [illegible] DEPTH	MIN	.027	.029	.031	.351	.353
J- HEAD SIDE HEIGHT	MIN	.221	.337	.394	.450	.562
J- SOCKET WIDTH ACROSS FLATS	MAX MIN	.2530 .2520	.3150 .3125	.3780 .3750	.3780 .3750	.5050 .5020
T- KEY ENGAGEMENT	MIN	.151	.182	.213	.245	.307
G- WALL THICKNESS	MIN	.102	.120	.140	.153	.192
F- RADIUS OR CHAMFER	MAX	.008	.008	.010	.010	.010
R- RADIUS	MAX MIN	.017 .012	.020 .015	.023 .018	.025 .020	.032 .026
E- HOLE LOCATION	MAX MIN	.104 .084	.123 .103	.141 .121	.160 .140	.198 .178
F- HOLE DIAMETER	MAX MIN	.053 .046	.067 .061	.067 .061	.067 .061	.067 .061
N- MINIMUM THREAD LENGTH		1.125	1.250	1.375	1.500	1.750
RECOMMENDED WIRE DIAMETER	MAX MIN	.040 .032	.057 .045	.057 .045	.057 .045	.057 .045
*DRILL HOLE ALIGNMENT PLUG		.030				
*MINIMUM BREAKING STRENGTH		8,190 LB	12,100 LB	16,700 LB	22,300 LB	[illegible] LB

L- LENGTH	PART NO.	PART NO.	PART NO.	PART NO.	PART NO.
3/8	MS 24676 -46				
1/2	-47	MS 24676 -62	MS 24676 -80	MS24676 -96	
5/8	-48	-63	-81	-97	
3/4	-49	-64	-82	-98	
7/8	-50	-65	-83	-99	
1	-51	-66	-84	-100	MS24676-116
1-1/4	-52	-67	-85	-101	-117
1-1/2	-53	-68	-86	-102	-118
1-3/4	-54	-69	-87	-103	-119
2	-55	-70	-88	-104	-120
2-1/4	-56	-71	-89	-105	-121
2-1/2	-57	-72	-90	-106	-122
2-3/4	-58	-73	-91	-107	-123
3	-59	-74	-92	-108	-124
3-1/4	-60	-75	-93	-109	-125
3-1/2	MS 24676 -61	-76	-94	-110	-126
4		-77	MS 24676 -95	-111	-127
4-1/2		-78		-112	-128
5		MS24676-79		-113	-129
5-1/2				-114	-130
6				MS24676-115	MS24676-131

Ⓑ CANCELLED

AFTER 7 JULY 1975 USE MS 24677

SEE SHEET 4

FSC 5305

APPROVED 7 APRIL 1960 REVISED Ⓑ FOR CHANGES SEE SHEETS 1, 2, 3, 4

P.A. Navy

Other Cust ORD AMD

PROCUREMENT SPECIFICATION FF-S-86

TITLE SCREW, CAP - SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, UNCOATED (UNC-3A)

SUPERSEDES

MILITARY STANDARD

MS 24676

SHEET 2

[illegible] Department of Defense [illegible]

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FED. SUP. CLASS 5305

NOMINAL SIZE		3/4	7/8	1		
THREADS PER INCH		10UNC-3A	9UNC-3A	8UNC-3A		
D - BODY DIAMETER	MAX MIN	.7500 .7405	.8750 .8647	1.0000 .9935		
A - HEAD DIAMETER	MAX MIN	1.125 1.107	1.312 1.291	1.500 1.479		
H - HEAD HEIGHT	MAX MIN	.750 .740	.875 .865	1.000 .988		
[illegible]	MIN	.446	.456	.455		
HEAD SIDE HEIGHT	MIN	.375	[illegible]	.904		
J - SOCKET WIDTH ACROSS FLATS	MAX MIN	.6312 .6250	.7573 .7500	.7573 .7500		
[illegible]	MIN	.375	.416	.47		
[illegible]	MIN	.245	.276	.320		
RADIUS OR CHAMFER	MAX	.013	.015	.016		
[illegible]	MAX MIN	.039 .030	.044 .034	.050 .040		
[illegible]	MAX MIN	.235 .215	.273 .253	.310 .290		
[illegible]	MAX MIN	.097 .052	.097 .081	.097 .081		
[illegible] THREAD LENGTH		2.000	2.250	2.500		
[illegible] DIAMETER	MAX MIN	.090 .064	.090 .064	.090 .064		
[illegible] STRENGTH		38,900 LB	53,000 LB	69,800 LB		

L - LENGTH	PART NO.		PART NO.		PART NO.					
1-1/4	MS24676-132									
1-1/2	-133									
1-3/4	-134									
2	-135		MS24676-147		MS24676-159					
2-1/4	-136		-148		-160					
2-1/2	-137		-149		-161					
2-3/4	-138		-150		-162					
3	-139		-151		-163					
3-1/4	-140		-152		-164					
3-1/2	-141		-153		-165					
4	-142		-154		-166					
4-1/2	-143		-155		-167					
5	-144		-156		-168					
5-1/2	-145		-157		-169					
6	MS24676-146		MS24676-158		MS24676-170					

⑤ CANCELLED

AFTER 7 JULY 1975 USE MS 24677

SEE SHEET 4

APPROVED 7 APRIL 1960 REVISED ⑤ FOR CHANGES SEE SHEETS 1, 2, 3, 4

TITLE

SCREW, CAP-SOCKET HEAD, HEXAGON, DRILLED SOCKET STEEL, UNCOATED (UNC-3A)

MILITARY STANDARD

MS24676

DD FORM 672-1

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FED. SUP. CLASS
5305

*Alignment Plug shall pass completely through the head without deflection.

**Based on 160,000 PSI minimum tensile strength. Breaking strengths are determined by the stress areas indicated in Screw-Thread Standards for Federal Services Handbook H-28.

MATERIAL: Steel, Alloy: Federal Standard 66.

HARDNESS: Rockwell, C36-43

PROTECTIVE COATING: Uncoated.

THREADS: The threads shall be in accordance with Specification MIL-S-7742

THREAD LENGTH-A: Screws which have a body length less than the minimum thread length shall be threaded within two thread pitches of the head.

LENGTH TOLERANCE:

LENGTH	DIAMETER	TOLERANCE
1/4 thru 1.000	#4 thru 1.000	±1/64
1-1/4 thru 2.000	#4 thru 1.000	+1/32 -1/64
2-1/4 thru 6.000	#4 thru 1.000	±1/32

DRILLED HOLE DATA:
(a) Screw sizes No. 4 and No. 6 shall have two (2) drilled holes spaced 180° and size No. 8 and above shall have six (6) drilled holes spaced 60°
(b) Edges of the holes on the outside of the head shall be chamfered 45° by .010 to .020 deep.
(c) On screw sizes below No. 8 the drilled hole shall be centered as close as practicable on the flat of the hexagon socket.
(d) On screw sizes No. 8 and above the drilled holes shall be within the flats of the hexagon socket and shall not break through the corners of the hexagon socket.

HEAD: Plain or knurled heads are optional.

DIMENSIONS: In inches unless otherwise specified.

NOTES:
(a) This Standard takes precedence over documents referenced herein.
(b) Reference documents shall be of the issue in effect on date of invitations for bids.

PART NUMBER = (MS Number) - (Screw dash number)
EXAMPLE = MS 24676 -1 is the part number for a screw #4-40NC-3A by 1/4 long Alloy Steel uncoated.

Ⓑ CANCELLED

AFTER 7 JULY 1976 USE MS 24677

INTERCHANGEABILITY RELATIONSHIP — MS 24677 SCREWS CAN REPLACE THE CANCELLED SCREWS OF THE SAME DASH NUMBERS FOR MOST APPLICATIONS, EXCEPT THOSE IN WHICH CADMIUM PLATED STEEL SCREWS ARE UNACCEPTABLE DUE TO TEMPERATURE, CORROSION, OR TOXICITY CONSIDERATIONS. THE CANCELLED MS 24676 SCREWS CANNOT ALWAYS REPLACE MS 24677 SCREWS AND SHOULD BE USED UNTIL EXISTING STOCK IS DEPLETED. USE ONLY SUPERSEDING MS SCREWS OF THE SAME DASH NUMBERS FOR DESIGN AND REPLACEMENT.

P.A. ...
Other Cust
ORD
ARM

PROCUREMENT SPECIFICATION
FF-S-86

TITLE
SCREW, CAP - SOCKET HEAD, HEXAGON, DRILLED
ALLOY STEEL, UNCOATED
(UNC-3A)

SUPERSEDES

MILITARY STANDARD
MS 24676

APP'D 7 APRIL 1966 REVISED Ⓑ FOR CHANGES SEE SHEETS 1, 2, 3, 4

DD FORM 1 OCT 62 672-1